

Non-Equilibrium Statistical Physics

Course notes (C. Jarzynski)

Francisco Gómez Serrano

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1 Thermodynamics, ensembles and ergodicity

1.1 Thermodynamics

Equilibrium is static; non-equilibrium means change in time, which is **non-universal**: it depends on the details. There are three levels of description: all particles; some particles “in a bath”; transitions between states.

Statement (R. Clausius, 1865). 1. The energy of the universe is constant.
2. The entropy of the universe tends toward a maximum.

- **Equilibrium states**: “nothing happens when left alone”.
- **State function**: an observable with a well-defined value in equilibrium, e.g. the internal energy $U = U(\text{state})$ and the entropy $S = S(\text{state})$.
- **Thermodynamic process**: begins and ends in equilibrium, e.g. the free expansion of a gas from state A to state B .
- **Reversible process**: the system is in equilibrium with the environment at all times. It is an idealisation, proceeds slowly, and can be reversed.

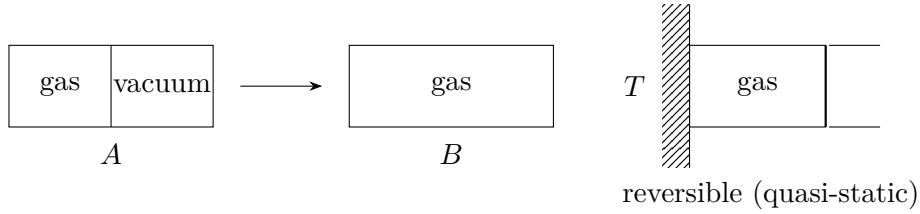


Figure 1.1: Free expansion from A to B (irreversible), and a gas compressed slowly by a piston in contact with a reservoir at temperature T (reversible).

First law.

$$\Delta U = U_B - U_A = W + Q, \quad dU = dW + dQ,$$

where W is the work performed *on* the system and Q the heat absorbed by the system; dU is exact while dW and dQ depend on the path.

Example: rubber band. A rubber band of length L in air at temperature T : (L, T) specify its equilibrium state, and

$$dU = \mathcal{J} dL + dQ \simeq 0 \quad \text{at fixed } T,$$

where $\mathcal{J}$ is the tension: work done on the band leaves as heat.

Clausius inequality.

$$\int_A^B \frac{dQ}{T} \leq \Delta S = S_B - S_A,$$

with equality for reversible and strict inequality for irreversible processes. For a reversible path, $\int_A^B dQ/T = S_B - S_A$ for any reversible path. For an isolated system $0 \leq \Delta S$.

For a system in contact with a single reservoir at temperature T ,

$$\frac{1}{T} \int_A^B dQ \leq \Delta S \implies Q_{A \rightarrow B} \leq T \Delta S \implies \Delta U - W_{A \rightarrow B} \leq T \Delta S,$$

so

$$W_{A \rightarrow B} \geq \Delta U - T \Delta S = \Delta F, \quad F := U - TS,$$

with F the **Helmholtz free energy**:

$$W_{A \rightarrow B} \geq \Delta F, \quad (1.1)$$

the second law of thermodynamics in the form of the **minimal work principle**.

Kelvin–Planck statement. For one reservoir and a cyclic process $A \rightarrow A$: $W \geq 0$, $Q = -W \leq 0$. “There is no process whose sole effect is the complete conversion of heat into work.”

Carnot statement (1824). A bound on the efficiency of heat engines (historically the first formulation of the second law).

1.2 Statistical ensembles and dynamics

A **statistical ensemble** is a probability distribution over microscopic states. For N particles in a box the microscopic state is

$$x = (\vec{q}_1, \dots, \vec{q}_N, \vec{p}_1, \dots, \vec{p}_N) \in \mathbb{R}^{6N},$$

described by a density $f(x)$ in phase space, with Hamiltonian $H(x)$; the surface $H(x) = E$ is the **energy shell**.

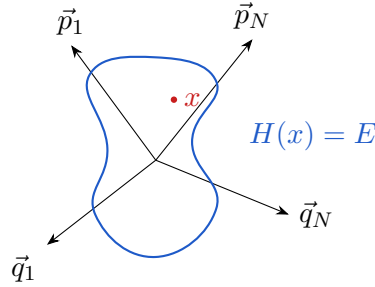


Figure 1.2: Phase space and the energy shell $H(x) = E$.

$$\Omega(E) = \int d^{6N}x \, \theta(E - H(x)), \quad \Sigma(E) = \frac{\partial \Omega}{\partial E} = \int d^{6N}x \, \delta(E - H(x)),$$

the volume of phase space enclosed by the energy shell and its derivative.

- **Microcanonical ensemble:** $\pi_E^\mu(x) = \frac{1}{\Sigma(E)} \delta(E - H(x))$.
- **Canonical ensemble:** $\pi_T^c(x) = \frac{1}{Z(\beta)} e^{-\beta H(x)}$, $\beta = 1/k_B T$.

Then

$$U = \begin{cases} \int d^{6N}x \, H(x) \pi_E^\mu(x) = E, \\ \int d^{6N}x \, H(x) \pi_T^c(x), \end{cases} \quad S = \begin{cases} k_B \ln \Sigma(E) - k_B \ln N!, \\ -k_B \int \pi^c \ln \pi^c - k_B \ln N!, \end{cases}$$

and the canonical distribution is a superposition of microcanonical ones,

$$\pi_T^c(x) = \frac{1}{Z} \int dE \, \Sigma(E) e^{-\beta E} \pi_E^\mu(x).$$

For an observable $A(x)$, $\langle A \rangle = \int \pi^\mu A$ or $\int \pi^c A$; if N is large enough (thermodynamic limit) both ensembles are equivalent.

Rubber band again. $\mathcal{J} = aT(L - L_0)$, $U = cT$, and the heat exchanged relaxes the temperature as $dQ/dt = -\alpha(T - T_0)$, with α^{-1} a time scale.

Types of dynamics. Hamiltonian dynamics; Langevin dynamics / Fokker–Planck equations; discrete-state dynamics.

Flows and the continuity equation. For a general dynamical system $\vec{x} = (x_1, \dots, x_r)$, $d\vec{x}/dt = \vec{v}(\vec{x})$, with trajectories $x_t = X(t)$ starting at x_0 , a density $f(\vec{x}, t)$ of trajectories obeys the **continuity equation**

$$\frac{\partial f}{\partial t} + \nabla \cdot (\vec{v}f) = 0.$$

Along a trajectory,

$$\frac{d}{dt}f(\vec{x}(t), t) = (\nabla f) \cdot \frac{d\vec{x}}{dt} + \frac{\partial f}{\partial t} = (\nabla f) \cdot \vec{v} - \nabla \cdot (\vec{v}f) = -(\nabla \cdot \vec{v})f,$$

so the divergence of the flow controls compression; a small volume ω transported along the flow changes as

$$\omega(\tau) = \omega(0) \exp \left(\int_0^\tau dt \nabla \cdot \vec{v}(x_t, t) \right).$$

Hamiltonian dynamics. For $x = (\vec{q}_1, \dots, \vec{p}_N) \in \mathbb{R}^{6N}$,

$$\frac{dq_i}{dt} = \frac{\partial H}{\partial p_i}, \quad \frac{dp_i}{dt} = -\frac{\partial H}{\partial q_i}; \quad \text{e.g.} \quad H = \frac{p^2}{2m} + V(q) \Rightarrow m \frac{d^2 q_i}{dt^2} = -\frac{\partial V}{\partial q_i}.$$

The velocity field is $\vec{v} = (\partial H / \partial p, -\partial H / \partial q)$, and

$$\frac{d}{dt}H(x(t)) = \sum_i \left(\frac{\partial H}{\partial q_i} \frac{dq_i}{dt} + \frac{\partial H}{\partial p_i} \frac{dp_i}{dt} \right) = \sum_i \left(\frac{\partial H}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial H}{\partial p_i} \frac{\partial H}{\partial q_i} \right) = 0$$

(energy conservation). Moreover

$$\nabla \cdot \vec{v} = \sum_i \left[\frac{\partial}{\partial q_i} \left(\frac{dq_i}{dt} \right) + \frac{\partial}{\partial p_i} \left(\frac{dp_i}{dt} \right) \right] = \sum_i \left(\frac{\partial^2 H}{\partial q_i \partial p_i} - \frac{\partial^2 H}{\partial p_i \partial q_i} \right) = 0 :$$

the density along trajectories does not change, i.e. the volume of phase space is preserved (**Liouville's theorem**), even if H depends on time (in which case $dH(x(t), t)/dt = \partial H / \partial t$). For the phase-space density,

$$\frac{\partial f}{\partial t} + \nabla \cdot (\vec{v}f) = \frac{\partial f}{\partial t} + \underbrace{(\nabla \cdot \vec{v})f}_0 + \vec{v} \cdot \nabla f = \frac{\partial f}{\partial t} + \sum_i \left(\frac{\partial H}{\partial p_i} \frac{\partial f}{\partial q_i} - \frac{\partial H}{\partial q_i} \frac{\partial f}{\partial p_i} \right) = 0,$$

the **Liouville equation**

$$\frac{\partial f}{\partial t} + \{f, H\} = 0.$$

Averages. For $A = A(x)$, the time average and the ensemble average are

$$\bar{A} := \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T dt A(x(t)), \quad \langle A \rangle_f := \int dx A(x) f(x).$$

Is $\langle A \rangle_\mu = \int dx \pi_E^\mu(x) A(x) \stackrel{?}{=} \bar{A}$?

1.3 Ergodicity and mixing in Hamiltonian dynamics

Consider an observable $A(q, p)$, explicitly independent of time (as is the Hamiltonian). Is $\bar{A} = \langle A \rangle_\mu$?

One degree of freedom. The energy shell $H(q, p) = E$ is a closed curve $\mathcal{E}$, traversed with period T . For a piece σ of the curve (obtained by cutting $\mathcal{E}$ with a region K of phase space), define the measure

$$\mu(\sigma) = \int_K dq dp \delta(E - H(q, p)) = \lim_{\delta E \rightarrow 0} \frac{|L|}{\delta E} \geq 0,$$

where L is the region between the shells E and $E + \delta E$ above σ ; in 1D it has units of time. Then $\mu(\mathcal{E}) = \Sigma(E) = T$.

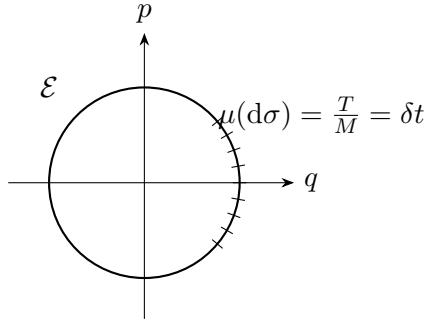


Figure 1.3: Dividing the energy curve into M cells of equal measure, each traversed in a time $\delta t = T/M$.

Dividing $\mathcal{E}$ into M cells $d\sigma_i$ of equal measure $\mu(d\sigma_i) = T/M = \delta t$, with points x_i ,

$$\int_{\mathcal{E}} d\mu A(x) = \lim_{M \rightarrow \infty} \sum_{i=1}^M \mu(d\sigma_i) A(x_i) = \int dq dp \delta(E - H(q, p)) A(q, p),$$

and

$$\begin{aligned} \bar{A} &= \frac{1}{T} \int_0^T dt A(q(t), p(t)) = \frac{1}{T} \lim_{M \rightarrow \infty} \delta t \sum_i A(q_i, p_i) = \frac{1}{\Sigma(E)} \lim_{M \rightarrow \infty} \sum_i \mu(d\sigma_i) A(q_i, p_i) \\ &= \frac{1}{\Sigma(E)} \int_{\mathcal{E}} d\mu A(x) = \frac{\int dq dp \delta(E - H) A}{\Sigma(E)}, \end{aligned}$$

that is,

$$\bar{A} = \frac{\int_{\mathcal{E}} d\mu A(x)}{\int_{\mathcal{E}} d\mu} = \langle A \rangle_\mu.$$

In 1D, time and microcanonical averages agree. (If the energy shell has several disconnected pieces, e.g. a double-well potential below the barrier, the trajectory only explores one of them.)

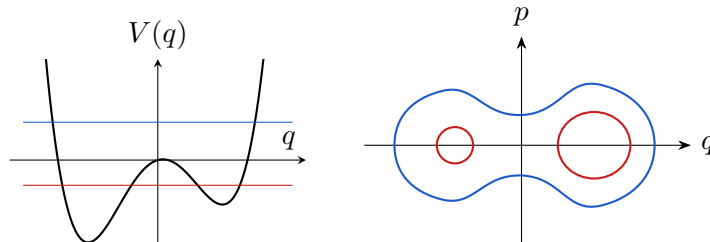


Figure 1.4: Double-well potential: above the barrier (blue) the energy curve is connected; below it (red) it splits into two components.

Many degrees of freedom. With $\Omega(E) = \int \theta(E - H)$, $\Sigma(E) = d\Omega/dE = \int \delta(E - H)$ and $\langle A \rangle_\mu = \frac{1}{\Sigma} \int A \delta(E - H)$, define for a piece σ of the energy shell $\mathcal{E}$ (cut out by a region K)

$$\mu(\sigma) := \int_K \delta(E - H(x)) = \lim_{\delta E \rightarrow 0} \frac{|L|}{\delta E}.$$

By Liouville's theorem a piece σ_0 evolves into σ_t with $\mu(\sigma_0) = \mu(\sigma_t)$. Dividing $\mathcal{E}$ into M cells,

$$\int_{\mathcal{E}} d\mu A(x) = \lim_{M \rightarrow \infty} \sum_{i=1}^M \mu(d\sigma_i) A(x_i) = \int \delta(E - H(x)) A(x),$$

and for $A = 1$, $\int_{\mathcal{E}} d\mu = \Sigma(E)$, so

$$\frac{\int_{\mathcal{E}} d\mu A}{\int_{\mathcal{E}} d\mu} = \frac{1}{\Sigma(E)} \int \delta(E - H(x)) A = \langle A \rangle_\mu.$$

The ergodic property. For (almost) any trajectory that starts at A , the trajectory will pass arbitrarily close to any point B of the energy shell, i.e. through any region σ with $\mu(\sigma) > 0$ (the exceptions, e.g. A on a periodic orbit, have measure zero). If the dynamics is ergodic, then $\overline{A} = \langle A \rangle_\mu$. In general the energy shell may split into several **ergodic components**.

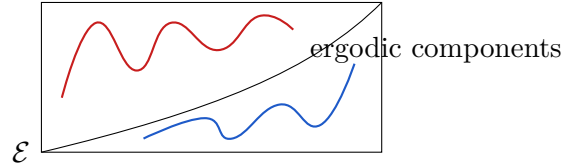


Figure 1.5: An energy shell split into two ergodic components.

The microcanonical density $f_\mu(x) = 1/\Sigma$ on $\mathcal{E}$ is stationary. If the shell splits into two invariant regions, $f(x) = C_1$ for $x \in 1$ and C_2 for $x \in 2$ is also stationary: the dynamics is then not ergodic. Dividing $\mathcal{E}$ into K cells $\{\eta_1, \dots, \eta_K\}$ of equal measure, $\mu(\eta_i) = \Sigma(E)/K$, a region σ_0 evolves into $\sigma_1, \sigma_2, \dots$ of the same measure but increasingly distorted shape.

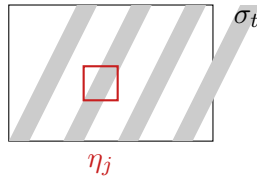


Figure 1.6: After a long time the region σ_t is stretched into thin filaments that fill the energy shell uniformly.

The mixing property. For every pair of regions,

$$\lim_{t \rightarrow \infty} \frac{\mu(\sigma_t \cap \eta_j)}{\mu(\eta_j)} = \frac{\mu(\sigma_t)}{\mu(\mathcal{E})} :$$

the evolved region ends up occupying every cell in proportion to its size. Mixing implies ergodicity, but ergodicity does not imply mixing.

Example: the baker's map. The unit square is squeezed vertically by a factor 2 and stretched horizontally to $[0, 2] \times [0, \frac{1}{2}]$; the right half is cut and placed on top of the left half. After n iterations a region is cut into 2^n thin horizontal strips spread over the whole square: the map is mixing.

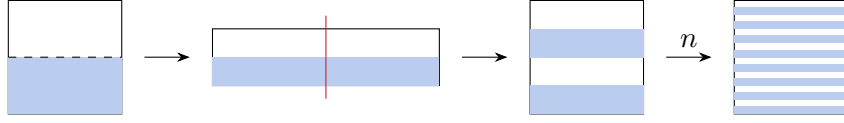


Figure 1.7: Baker's map: stretch, cut and stack. A region becomes 2^n strips.

For two regions A, B of equal measure, after a long time each small cell η contains equal amounts of the evolved images of A and B : with mixing, the system “loses memory in some sense”.

1.4 Relaxation, correlations and chaotic billiards

For mixing dynamics, any observable $A(x)$ and any initial density $f(x, t)$ on the energy shell,

$$\lim_{t \rightarrow \infty} \int_{\mathcal{E}} d\mu f(x, t) A(x) = \langle A \rangle_{\mu} = \int_{\mathcal{E}} d\mu \frac{A(x)}{\Sigma} :$$

effectively $f(x, t) \xrightarrow{t \rightarrow \infty} 1/\Sigma(E)$ (in a coarse-grained sense). Indeed, start from

$$f(x, 0) = \begin{cases} 1/\mu(\sigma_0), & x \in \sigma_0, \\ 0, & x \notin \sigma_0, \end{cases} \quad f(x, t) = \begin{cases} 1/\mu(\sigma_t), & x \in \sigma_t, \\ 0, & x \notin \sigma_t. \end{cases}$$

The average of f over a cell η is

$$\frac{\int_{\eta} f(x, t) d\mu}{\int_{\eta} d\mu} = \frac{\int_{\eta \cap \sigma_t} d\mu / \mu(\sigma_0)}{\mu(\eta)} = \frac{\mu(\eta \cap \sigma_t)}{\mu(\eta)\mu(\sigma_t)} \xrightarrow{t \rightarrow \infty} \frac{\mu(\sigma_t)}{\mu(\mathcal{E})} \frac{1}{\mu(\sigma_t)} = \frac{1}{\mu(\mathcal{E})},$$

using $\mu(\sigma_t) = \mu(\sigma_0)$.

Time correlation functions. For $A(x)$ with $\bar{A} = \langle A \rangle_{\mu}$, let $\delta A(x) := A(x) - \bar{A}$ and

$$C_A(t) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T ds \delta A(x(s)) \delta A(x(s+t)) = \langle \delta A \delta A(t) \rangle_{\mu} = \langle A A_t \rangle - \langle A \rangle^2.$$

For mixing dynamics $C_A(t) \xrightarrow{t \rightarrow \infty} \langle A \rangle \langle A_t \rangle - \langle A \rangle^2 = 0$, while $C_A(0) = \langle A^2 \rangle - \langle A \rangle^2$. $C_A(t)$ decays over a correlation time t_c .

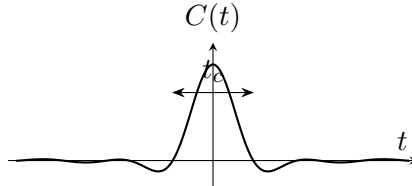


Figure 1.8: A time correlation function decaying over a correlation time t_c .

Chaotic billiards. Sinai's billiard (1963: a square with a circular obstacle) and Bunimovich's stadium (1979) are ergodic and mixing.

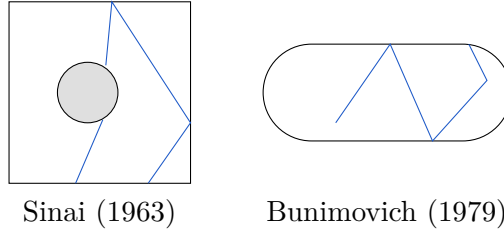


Figure 1.9: Chaotic billiards.

Statement (Boltzmann–Sinai hypothesis). A sufficiently dilute gas of $N \geq 2$ hard spheres in 2 or 3 dimensions is ergodic and mixing.

Maxwell distribution from the microcanonical ensemble. For $N \gg 1$ particles in $d = 3$ in a volume V , $E = \sum_{i=1}^N |\vec{p}_i|^2 / 2m$, i.e. $\sum_i |\vec{p}_i|^2 = 2mE$. With $\vec{\zeta}_i = \vec{p}_i / \sqrt{2mE}$ the momenta lie on the unit hypersphere in $3N$ dimensions, and dilating,

$$\begin{aligned} \Omega(E) &= V^N \int d^{3N} p \, \theta\left(E - \sum_i \frac{p_i^2}{2m}\right) = V^N (2mE)^{3N/2} \int d^{3N} z \, \theta(1 - z^2) \\ &= V^N (2mE)^{3N/2} \frac{\pi^{3N/2}}{\Gamma(\frac{3N}{2} + 1)}, \end{aligned}$$

with $\Gamma(k) = \int_0^\infty e^{-t} t^{k-1} dt = (k-1)!$ for $k \in \mathbb{N}$. Hence

$$\Sigma(E) = V^N \frac{(2\pi m)^k E^{k-1}}{\Gamma(k)}, \quad k = \frac{3N}{2}.$$

Now write $\vec{P} = (P_1, \dots, P_{3N}) = (\vec{p}_1, \dots, \vec{p}_N) = (P_1, \vec{P}')$. The distribution of a single momentum component is

$$\begin{aligned} \rho(P_1) &\propto \int d^{3N-1} P' \, \delta\left(E - \frac{P_1^2}{2m} - \frac{|\vec{P}'|^2}{2m}\right) = \frac{\partial}{\partial E} \int d^{3N-1} P' \, \theta\left(E - \frac{P_1^2}{2m} - \frac{|\vec{P}'|^2}{2m}\right) \\ &= \frac{\partial}{\partial E} [\text{volume of a } (3N-1)\text{-d hypersphere of radius } \sqrt{2mE - P_1^2}] \propto \frac{\partial}{\partial E} (2mE - P_1^2)^{\frac{3N-1}{2}} \\ &\propto \left(1 - \frac{P_1^2}{2mE}\right)^{\frac{3N-3}{2}} \\ &= \left(1 - \frac{P_1^2}{3mNk_B T}\right)^{\frac{3N-3}{2}} = \left[\left(1 - \frac{P_1^2}{3mNk_B T}\right)^N\right]^{\frac{3}{2} - \frac{3}{2N}}, \end{aligned}$$

using $E = \frac{3}{2} N k_B T$. Since $\lim_{N \rightarrow \infty} (1 - y/N)^N = e^{-y}$,

$$\rho(P_1) \xrightarrow{N \rightarrow \infty} e^{-P_1^2 / 2m k_B T},$$

the **Maxwell distribution**. Boltzmann instead used the **Stosszahlansatz** (molecular chaos hypothesis).

2 Stochastic dynamics and relaxation

2.1 Diffusive dynamics

Diffusive dynamics is described by Langevin equations and the Fokker–Planck equation. Consider a $\sim 1\text{ }\mu\text{m}$ particle in water, in a potential $V(x) = \frac{1}{2}\kappa x^2$:

$$m\ddot{x} = -\gamma\dot{x} + [\text{noise}] - \kappa x,$$

with $[\gamma] = \text{mass/time}$ and $[\kappa] = \text{mass/time}^2$. There are two time scales:

$$t_p = \frac{m}{\gamma} \text{ (relaxation of the velocity/momentum),} \quad t_\kappa = \frac{\gamma}{\kappa} \text{ (relaxation of the position).}$$

For $t_p \ll t_\kappa$ (**overdamped limit**) we don't care about the momentum of the particle.

Random velocity. Consider $dx/dt = \tilde{v}(t)$, where $\tilde{v}(t)$ is statistically stationary, $\langle \tilde{v}(t) \rangle = 0$, and its correlation $C_{\tilde{v}}(t) := \langle \tilde{v}(s)\tilde{v}(s+t) \rangle$ decays to zero faster than $1/t$. With $\Delta x = \int_0^{\Delta t} \tilde{v}(t) dt$, $\langle \Delta x \rangle = 0$ and

$$\begin{aligned} \langle \Delta x^2 \rangle &= \int_0^{\Delta t} dt' \int_0^{\Delta t} dt'' \langle \tilde{v}(t')\tilde{v}(t'') \rangle \\ &= \int_0^{\Delta t} dt' \int_0^{\Delta t} dt'' C_{\tilde{v}}(t' - t'') \\ &= \int_{-\Delta t}^{\Delta t} dt C_{\tilde{v}}(t)(\Delta t - |t|) \\ &= \Delta t \int_{-\Delta t}^{\Delta t} dt C_{\tilde{v}}(t) \left(1 - \frac{|t|}{\Delta t}\right), \end{aligned}$$

changing variables to $t = t' - t''$. For $\Delta t \gg t_c$ the triangular factor is $\simeq 1$ wherever $C_{\tilde{v}}$ is non-negligible, so

$$\lim_{\Delta t \rightarrow \infty} \frac{\langle \Delta x^2 \rangle}{\Delta t} = \int_{-\infty}^{\infty} dt C_{\tilde{v}}(t) =: 2D.$$

An ensemble of trajectories starting at x_0 spreads as

$$f(x, t) = \frac{1}{\sqrt{4\pi Dt}} e^{-(x-x_0)^2/4Dt}, \quad \langle (x - \langle x \rangle)^2 \rangle = 2Dt.$$

White noise and the diffusion equation. In the limit of short correlations, $\langle \tilde{v}(t) \rangle = 0$ and $\langle \tilde{v}(s)\tilde{v}(s+t) \rangle = 2D\delta(t)$. If $f(x, 0) = \delta(x - x_0)$, then $f(x, t) = \frac{1}{\sqrt{4\pi Dt}} e^{-(x-x_0)^2/4Dt}$, which solves the **diffusion equation**

$$\frac{\partial f}{\partial t} = D \frac{\partial^2 f}{\partial x^2}.$$

The **kernel** or **propagator**

$$K(x, t | x_0, t_0) = \frac{1}{\sqrt{4\pi D(t - t_0)}} e^{-\frac{(x-x_0)^2}{4D(t-t_0)}}$$

evolves an arbitrary $f(x, t_0)$: for $t > t_0$, $f(x, t) = \int dx_0 f(x_0, t_0) K(x, t | x_0, t_0)$.

Drift. For $\dot{x} = v + \tilde{v}(t)$,

$$\frac{\partial f}{\partial t} = -v \frac{\partial f}{\partial x} + D \frac{\partial^2 f}{\partial x^2} :$$

the distribution drifts and spreads.

Ornstein–Uhlenbeck process. For $\dot{x} = -\alpha x + \tilde{v}(t)$ with $C_{\tilde{v}}(t) = 2D\delta(t)$,

$$\frac{\partial f}{\partial t} = \alpha \frac{\partial}{\partial x}(xf) + D \frac{\partial^2 f}{\partial x^2},$$

a competition between the restoring drift and diffusion. Starting from $f(x, 0) = \delta(x - x_0)$,

$$f(x, t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad \mu(t) = x_0 e^{-\alpha t}, \quad \sigma^2(t) = \frac{D}{\alpha} (1 - e^{-2\alpha t}),$$

which relaxes to the stationary distribution

$$\pi(x) = \sqrt{\frac{\alpha}{2\pi D}} e^{-\alpha x^2/2D}, \quad C_x(t) = \frac{D}{\alpha} e^{-\alpha|t|} = \int dx \int dx_0 \pi(x_0) K(x, t | x_0, 0) x x_0 = \langle x_t x_0 \rangle.$$

1. Overdamped harmonic oscillator. Neglecting inertia, $\gamma\dot{x} = -\kappa x$, i.e. $\dot{x} = -\frac{\kappa}{\gamma}x$ ($t_\kappa = \gamma/\kappa$). Adding noise,

$$\dot{x} = -\frac{\kappa}{\gamma}x + \xi(t), \quad \langle \xi(s)\xi(s+t) \rangle = 2D\delta(t),$$

so mathematically $\pi(x) \propto e^{-\kappa x^2/2\gamma D}$, while physically $\pi(x) \propto e^{-\beta\kappa x^2/2}$. Therefore

$$\beta\gamma D = 1,$$

the **Einstein–Smoluchowski relation**, a fluctuation–dissipation relation.

2. Free Brownian particle, not overdamped.

$$\dot{p} = -\gamma \frac{p}{m} + \xi(t), \quad \langle \xi(s)\xi(s+t) \rangle = 2D_p\delta(t), \quad \frac{\partial f}{\partial t} = \frac{\gamma}{m} \frac{\partial}{\partial p}(pf) + D_p \frac{\partial^2 f}{\partial p^2},$$

gives $\pi(p) \propto e^{-\gamma p^2/2mD_p}$ ($t_p = m/\gamma$). Physically $\pi(p) \propto e^{-\beta p^2/2m}$, so

$$\gamma = \beta D_p,$$

the momentum-space analogue. The momentum is a random signal with finite correlation time,

$$C_p(t) = \frac{mD_p}{\gamma} e^{-\gamma|t|/m}, \quad C_p(0) = \frac{mD_p}{\gamma} = mk_B T = \langle p^2 \rangle.$$

Then $\dot{x} = p/m = \tilde{v}(t)$ with $C_{\tilde{v}}(t) = C_p(t)/m^2$, and $\langle \Delta x^2 \rangle = 2D\Delta t$ for $\Delta t \gg t_p$ with

$$2D = \int_{-\infty}^{\infty} C_{\tilde{v}}(t) dt = \frac{1}{m^2} \int_{-\infty}^{\infty} C_p(t) dt = \frac{2}{m\beta} \int_0^{\infty} dt e^{-\gamma t/m} = \frac{2}{\beta\gamma} \implies \beta\gamma D = 1.$$

Overdamped Brownian particle in an arbitrary potential.

$$\dot{x} = -\frac{1}{\gamma} \frac{\partial V}{\partial x} + \xi(t), \quad \langle \xi(s)\xi(s+t) \rangle = 2D\delta(t),$$

$$\frac{\partial f}{\partial t} = \frac{1}{\gamma} \frac{\partial}{\partial x} \left(\frac{\partial V}{\partial x} f \right) + D \frac{\partial^2 f}{\partial x^2} = D \frac{\partial}{\partial x} \left[e^{-V/\gamma D} \frac{\partial}{\partial x} (e^{V/\gamma D} f) \right],$$

so $\pi(x) \propto e^{-V(x)/\gamma D}$ (maths) and $\pi(x) \propto e^{-\beta V(x)}$ (physics), giving again $\beta\gamma D = 1$.

Underdamped particle in a potential. With $H(x, p) = \frac{p^2}{2m} + V(x)$,

$$\dot{x} = \frac{p}{m}, \quad \dot{p} = -\frac{\partial V}{\partial x} - \frac{\gamma}{m}p + \xi(t), \quad \langle \xi(s)\xi(s+t) \rangle = 2D_p\delta(t),$$

and the distribution $f(x, p, t)$ obeys

$$\frac{\partial f}{\partial t} = -\frac{p}{m} \frac{\partial f}{\partial x} + \frac{\partial V}{\partial x} \frac{\partial f}{\partial p} + \frac{\gamma}{m} \frac{\partial}{\partial p}(pf) + D_p \frac{\partial^2 f}{\partial p^2} = \{H, f\} + D_p \frac{\partial}{\partial p} \left[e^{-\gamma p^2/2mD_p} \frac{\partial}{\partial p} (e^{\gamma p^2/2mD_p} f) \right],$$

whose stationary solution is

$$\pi(x, p) \propto e^{-\beta H(x, p)} \quad \text{iff} \quad \frac{\gamma}{D_p} = \beta.$$

2.2 Langevin and Fokker–Planck equations

A **Langevin equation** $\dot{x} = v(x) + [\text{white noise}]$ corresponds to the **Fokker–Planck equation** for $f(x, t)$,

$$\frac{\partial f}{\partial t} = -\frac{\partial}{\partial x}(af) + \frac{\partial^2}{\partial x^2}(bf) =: \hat{\mathcal{L}}f, \quad a = a(x, t), \quad b = b(x, t) \geq 0.$$

For an observable $A(x)$, with $\langle g_1 | g_2 \rangle := \int dx \, g_1^*(x) g_2(x)$,

$$\begin{aligned} \frac{d}{dt} \langle A \rangle_t &= \int dx \, A(x) \hat{\mathcal{L}}f = \int dx \, A(x) \left[-\frac{\partial}{\partial x}(af) + \frac{\partial^2}{\partial x^2}(bf) \right] \\ &= \underbrace{\int dx \, \frac{\partial}{\partial x} \left\{ A \left(-af + \frac{\partial}{\partial x}(bf) \right) \right\}}_{0 \text{ (boundary term)}} \\ &\quad - \int dx \, \frac{\partial A}{\partial x} \left(-af + \frac{\partial}{\partial x}(bf) \right) = \int dx \, \left[\frac{\partial A}{\partial x} a + \frac{\partial^2 A}{\partial x^2} b \right] f, \end{aligned}$$

i.e.

$$\frac{d}{dt} \langle A \rangle_t = \int dx \, (\hat{\mathcal{L}}^\dagger A) f = \langle \hat{\mathcal{L}}^\dagger A | f \rangle = \langle A | \hat{\mathcal{L}} | f \rangle, \quad \hat{\mathcal{L}}^\dagger := a \frac{\partial}{\partial x} + b \frac{\partial^2}{\partial x^2},$$

the **adjoint** of $\hat{\mathcal{L}}$. For time-independent $a = a(x)$, $b = b(x)$, the probability distribution after a long time is stationary, and

$$\frac{d}{dt} |f_t\rangle = \hat{\mathcal{L}} |f_t\rangle \implies |f_t\rangle = e^{\hat{\mathcal{L}}t} |f_0\rangle = \sum_{n=0}^{\infty} \frac{t^n}{n!} \hat{\mathcal{L}}^n |f_0\rangle.$$

Defining $A(x, t)$ by $A(x, 0) = A(x)$ and

$$\frac{\partial A(x, t)}{\partial t} = \hat{\mathcal{L}}^\dagger A(x, t) \quad (\text{Kolmogorov backward equation}), \quad |A_t\rangle = e^{\hat{\mathcal{L}}^\dagger t} |A_0\rangle,$$

we have $\langle A \rangle_t = \langle A_0 | f_t \rangle = \langle A_t | f_0 \rangle$.

2.3 Spectral properties of the Fokker–Planck operator

Using these definitions,

$$\begin{aligned} \langle A \rangle_t &= \int dx \, A(x) f(x, t) = \langle A_0 | f_t \rangle = \langle A_0 | e^{\hat{\mathcal{L}}t} | f_0 \rangle \\ &= \langle e^{\hat{\mathcal{L}}^\dagger t} A_0 | f_0 \rangle = \langle A_t | f_0 \rangle = \int dx \, A(x, t) f(x, 0). \end{aligned}$$

(Compare quantum mechanics: $\hat{H} = \hat{H}^\dagger$, $i\hbar \partial_t \psi = \hat{H} \psi$, $\psi_t = e^{-i\hat{H}t/\hbar} \psi_0$; here $\hat{\mathcal{L}}$ is *not* self-adjoint.)

Stationary state. $\alpha_0(x) = 1$ satisfies $\hat{\mathcal{L}}^\dagger \alpha_0 = 0$. For $b(x) > 0$, the function

$$\varphi_0(x) = \pi(x) = \frac{1}{b(x)} e^{\psi(x)}, \quad \psi(x) = \int_{-\infty}^x dx' \frac{a(x')}{b(x')},$$

satisfies $\hat{\mathcal{L}}\pi = 0$, and if $\int dx \pi(x) < \infty$ it can be normalised, $\int dx \pi(x) = 1$.

Transient solutions. Suppose $\hat{\mathcal{L}}\varphi = \lambda\varphi$ and $\hat{\mathcal{L}}^\dagger\alpha = \lambda^*\alpha$ with $\lambda \neq 0$ (so that $\partial_t\varphi = \lambda\varphi$). Then

$$\lambda \int dx \varphi(x) = \lambda \langle 1 | \varphi \rangle = \langle 1 | \hat{\mathcal{L}}\varphi \rangle = \langle \hat{\mathcal{L}}^\dagger 1 | \varphi \rangle = 0,$$

so φ has zero integral (it takes both signs). For small ε , $f(x, 0) = \pi(x) + \varepsilon\varphi(x) \geq 0$ is a normalised distribution, and

$$f(x, t) = e^{\hat{\mathcal{L}}t}(\pi + \varepsilon\varphi) = \pi(x) + \varepsilon e^{\lambda t}\varphi(x) \implies \text{Re } \lambda < 0 \text{ if } \lambda \neq 0.$$

Hence the eigenvalues can be ordered as

$$0 = \lambda_0 > \text{Re } \lambda_1 \geq \text{Re } \lambda_2 \geq \dots,$$

with right eigenfunctions $\varphi_n(x) \rightarrow |n\rangle$ and left eigenfunctions $\alpha_n(x) \rightarrow \langle n|$:

$$\langle m | n \rangle = \int dx \alpha_m^*(x) \varphi_n(x) = \delta_{mn}.$$

2.4 Relaxation and relative entropy

For $\partial_t f = -\partial_x(af) + \partial_x^2(bf) =: \hat{\mathcal{L}}f$ with $\hat{\mathcal{L}}^\dagger = a\partial_x + b\partial_x^2$, $a = a(x)$, $b = b(x)$, the eigenstates satisfy $\hat{\mathcal{L}}\varphi = \lambda\varphi$ and $\langle g_1 | \hat{\mathcal{L}}g_2 \rangle = \langle \hat{\mathcal{L}}^\dagger g_1 | g_2 \rangle$; $\alpha_0 = 1$, $\varphi_0 = \pi$, $\langle 1 | \pi \rangle = \langle \alpha_0 | \varphi_0 \rangle = 1$.

Bi-orthogonality. If $\lambda_m \neq \lambda_n$,

$$\langle \alpha_n | \hat{\mathcal{L}} | \varphi_m \rangle = \lambda_m \langle \alpha_n | \varphi_m \rangle = \langle \hat{\mathcal{L}}^\dagger \alpha_n | \varphi_m \rangle = \lambda_n \langle \alpha_n | \varphi_m \rangle \implies \langle \alpha_n | \varphi_m \rangle = 0.$$

Assuming $\{\varphi_n\}$ and $\{\alpha_n\}$ are complete with $\langle \alpha_n | \varphi_m \rangle = \delta_{nm}$, any $f(x) = \sum_n c_n \varphi_n(x)$ has $c_m = \langle \alpha_m | f \rangle$, so

$$|f\rangle = \sum_n |\varphi_n\rangle \langle \alpha_n | f \rangle, \quad \sum_n |\varphi_n\rangle \langle \alpha_n| = \mathbb{K}.$$

For an initial distribution, $1 = \int dx f(x, 0) = \sum_n c_n \int dx \varphi_n(x) = c_0$, so

$$f(x, 0) = \pi(x) + \sum_{n=1}^{\infty} c_n \varphi_n(x), \quad f(x, t) = e^{\hat{\mathcal{L}}t} f(x, 0) = \pi(x) + \sum_{n=1}^{\infty} c_n e^{\lambda_n t} \varphi_n(x),$$

and $\lim_{t \rightarrow \infty} f(x, t) = \pi(x)$. Normalisation is conserved: $\frac{d}{dt} \int dx f = \int dx 1 \cdot \hat{\mathcal{L}}f = \langle \alpha_0 | \hat{\mathcal{L}}f \rangle = \langle \hat{\mathcal{L}}^\dagger \alpha_0 | f \rangle = 0$.

Example: double well. For $\partial_t f = \frac{1}{\gamma} \partial_x (V' f) + D \partial_x^2 f$, $\pi \propto e^{-\beta V}$, with peaks at the two minima. The slowest mode φ_1 is positive in one well and negative in the other, so at long times

$$f(x, t) \simeq \pi(x) + c_1 \varphi_1(x) e^{\lambda_1 t}$$

describes the slow transfer of probability between the wells over the barrier.

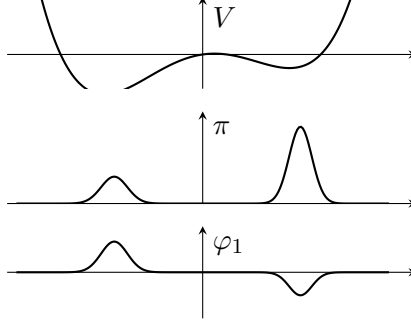


Figure 2.1: Double-well potential, stationary distribution $\pi = \varphi_0$ and the slowest relaxation mode φ_1 .

Relative entropy (Kullback–Leibler divergence). For probability distributions f, g with the same support,

$$\mathcal{D}(f|g) = \int f \ln \frac{f}{g} = - \int f \ln \frac{g}{f} \geq - \int f \left(\frac{g}{f} - 1 \right) = 0,$$

using $\ln x \leq x - 1$. $\mathcal{D}(f|g) \approx 0$ when $f \approx g$, $\mathcal{D} \gg 1$ for very different distributions, and in general $\mathcal{D}(f|g) \neq \mathcal{D}(g|f)$.

For overdamped diffusion with $\pi(x) = \frac{1}{Z} e^{-\beta V(x)}$, using $\hat{\mathcal{L}}^\dagger = -\frac{1}{\gamma} V' \partial_x + D \partial_x^2$ and $\beta \gamma D = 1$,

$$\begin{aligned} \frac{d}{dt} \mathcal{D}(f|\pi) &= \int dx \left(\dot{f} \ln \frac{f}{\pi} + f \frac{\dot{f}}{f} \right) = \int dx \ln \frac{f}{\pi} \hat{\mathcal{L}} f + \underbrace{\int \dot{f} dx}_0 = \int dx \left(\hat{\mathcal{L}}^\dagger \ln \frac{f}{\pi} \right) f \\ &= -D \int dx \frac{(f')^2}{f} - 2D\beta \int dx V' f' - \beta^2 D \int dx f (V')^2, \end{aligned}$$

so

$$\dot{\mathcal{D}}(f|\pi) = -D \int dx \frac{1}{f} (f' + \beta V' f)^2 \leq 0 :$$

the relative entropy is always decreasing, and f relaxes to π .

Driven systems. Now let $V = V(x, t)$: $\dot{x} = -\frac{1}{\gamma} \partial_x V + \xi(t)$. Along a trajectory,

$$\Delta V = V(x(\tau), \tau) - V(x(0), 0) = \int_0^\tau dt \frac{d}{dt} V(x(t), t) = \int_0^\tau dt \left[\underbrace{\frac{\partial V}{\partial t}}_{dW/dt} + \underbrace{\frac{\partial V}{\partial x} \frac{dx}{dt}}_{dQ/dt} \right].$$

With $V(x, \lambda)$, $\lambda = \lambda(t)$ driven from A to B for $t \in [0, \tau]$,

$$\Delta V = \underbrace{\int_0^\tau \dot{\lambda} \frac{\partial V}{\partial \lambda} dt}_W + \underbrace{\int_0^\tau \dot{x} \frac{\partial V}{\partial x} dt}_Q.$$

The equilibrium states are

$$\pi(x, \lambda) = \frac{1}{Z(\lambda)} e^{-\beta V(x, \lambda)} = e^{\beta(F_\lambda - V(x, \lambda))}, \quad F_\lambda = -\beta^{-1} \ln Z_\lambda,$$

and the system starts in $\pi_A(x) = \pi(x, A)$. During driving $f(x, t)$ lags behind $\pi_{\lambda(t)}$, and after $t > \tau$ it relaxes to π_B .

2.5 The second law and linear response

Let $\mathcal{D}(t) = \mathcal{D}(f(x, t) | \pi(x, \lambda(t))) = \int f \ln(f/\pi)$. Then

$$\dot{\mathcal{D}}(t) = \int \left(\dot{f} \ln \frac{f}{\pi} + \dot{f} - f \frac{\dot{\pi}}{\pi} \right), \quad \frac{\dot{\pi}}{\pi} = \frac{d}{dt} \ln \pi = \beta \dot{\lambda} \left(\frac{\partial F_\lambda}{\partial \lambda} - \frac{\partial V}{\partial \lambda} \right),$$

$$\begin{aligned} \dot{\mathcal{D}} &= -D \underbrace{\int \frac{1}{f} (f' + \beta f V')^2}_{\leq 0} - \beta \int \dot{\lambda} f \left(\frac{\partial F_\lambda}{\partial \lambda} - \frac{\partial V}{\partial \lambda} \right) \\ &\leq -\beta \left(\dot{\lambda} \frac{\partial F_\lambda}{\partial \lambda} - \dot{\lambda} \int f \frac{\partial V}{\partial \lambda} \right) = \beta (\langle \dot{W} \rangle - \dot{F}). \end{aligned}$$

Integrating, and using $\mathcal{D}(0) = 0$ (start in equilibrium) and $\mathcal{D}(\infty) = 0$ (final relaxation),

$$\int_0^\infty \dot{\mathcal{D}}(t) dt \leq \beta (\langle W \rangle - \Delta F) \implies \langle W \rangle \geq \Delta F,$$

the second law (1.1) on average. Stopping at $t = \tau$ instead,

$$\langle W \rangle \geq \Delta F + \beta^{-1} \underbrace{\mathcal{D}(f(x, \tau) | \pi_B(x))}_{\text{penalty}}.$$

With $\langle \Delta V \rangle = \langle W \rangle + \langle Q \rangle$,

$$\langle \Delta V \rangle - \langle Q \rangle \geq \Delta F \implies \langle Q \rangle \leq \langle \Delta V \rangle - \Delta F = \beta^{-1} \Delta S, \quad S_\lambda = - \int \pi_\lambda \ln \pi_\lambda,$$

i.e. $\langle Q \rangle / T \leq \Delta S$ (Clausius, with $k_B = 1$).

Linear response theory. Ohm's law, $\vec{J} = \sigma \vec{E}$, defines a **transport coefficient** σ .

Onsager's regression hypothesis (1931). Consider $\dot{x} = -\frac{1}{\gamma} \partial_x V + \xi(t)$ with $\langle x \rangle_0^{\text{eq}} = 0$ (assumption), and

$$V(x, t) = \begin{cases} V_0(x) - \varepsilon x, & t \leq 0, \\ V_0(x), & t > 0. \end{cases}$$

With $\pi_0(x) = \frac{1}{Z_0} e^{-\beta V_0(x)}$, the initial state is

$$f(x, 0) = \frac{1}{Z_\varepsilon} e^{-\beta V_0(x) + \beta \varepsilon x}, \quad Z_\varepsilon = \int dx e^{-\beta V_0} e^{\beta \varepsilon x} \simeq \int dx e^{-\beta V_0} (1 + \beta \varepsilon x) = Z_0.$$

For $t > 0$, $f(x, t) = \int dx' K(x, t | x', 0) f(x', 0)$, and

$$\begin{aligned} \langle x \rangle_t &= \int dx x f(x, t) = \int dx \int dx' x K(x, t | x', 0) \frac{1}{Z_0} e^{-\beta V_0(x')} (1 + \beta \varepsilon x') \\ &= \underbrace{\int dx x \pi_0(x)}_0 + \beta \varepsilon \int dx \int dx' x x' \pi_0(x') K(x, t | x', 0), \end{aligned}$$

$$\langle x \rangle_t = \beta \varepsilon C_x(t), \quad \langle x \rangle_0 = \beta \varepsilon \langle x^2 \rangle^{\text{eq}} = \beta \varepsilon C_x(0), \quad \frac{\langle x \rangle_t}{\langle x \rangle_0} = \frac{C_x(t)}{C_x(0)} :$$

the relaxation of a small macroscopic perturbation follows the equilibrium correlation function.

General linear response. Start in equilibrium ($t < 0$), perturb by $-\varepsilon\alpha(t)A(x)$ for $t \geq 0$, and observe $\langle B \rangle_{t>0}$ with $B = B(x, p)$, $\langle B \rangle^{\text{eq}} = 0$. With $\pi = \frac{1}{Z_0}e^{-\beta H_0(x, p)}$ and $H(x, p, t > 0) = H_0(x, p) - \varepsilon\alpha(t)A(x)$,

$$\frac{\partial f}{\partial t} = \{H, f\} + D_p \frac{\partial}{\partial p} \left(\pi \frac{\partial f}{\partial p} \right) = \hat{\mathcal{L}}_0 f - \varepsilon\alpha(t)\{A, f\}.$$

Expanding $f = f_0 + \varepsilon f_1 + \varepsilon^2 f_2 + \dots$:

- order ε^0 : $\partial_t f_0 = \hat{\mathcal{L}}_0 f_0 \Rightarrow f_0 = \pi(x, p) \propto e^{-\beta(p^2/2m + V(x))}$;
- order ε^1 :

$$\frac{\partial f_1}{\partial t} = \hat{\mathcal{L}}_0 f_1 - \alpha(t)\{A, f_0\} = \hat{\mathcal{L}}_0 f_1 - \alpha(t) \frac{\partial A}{\partial x} \frac{\partial f_0}{\partial p} = \hat{\mathcal{L}}_0 f_1 + \alpha(t) \beta \underbrace{\frac{\partial A}{\partial x} \frac{p}{m}}_{\dot{A}} \pi,$$

using $\partial_p f_0 = -\beta \frac{p}{m} \pi$. Hence

$$\begin{aligned} f_1(x, p, t) &= \int_0^t ds \alpha(s) \beta e^{(t-s)\hat{\mathcal{L}}_0} \dot{A}(x, p) \pi(x, p) \\ &= \int_0^t ds \alpha(s) \beta \int dx' dp' K_0(x, p, t-s | x', p', 0) \dot{A}(x', p') \pi(x', p'). \end{aligned}$$

With $\Gamma = (x, p)$,

$$\begin{aligned} \langle B \rangle_t &= \int d\Gamma f(\Gamma, t) B(\Gamma) = \varepsilon \int_0^t ds \alpha(s) \beta \int d\Gamma \int d\Gamma' B(\Gamma) K_0(\Gamma, t-s | \Gamma', 0) \dot{A}(\Gamma') \pi(\Gamma') \\ &= \varepsilon \int_0^t ds \alpha(s) \beta \left\langle \dot{A}(\Gamma_0) B(\Gamma_{t-s}) \right\rangle_0, \end{aligned}$$

that is,

$$\langle B \rangle_t = \varepsilon \int_0^t ds \alpha(s) R_{AB}(t-s), \quad R_{AB}(t) := \beta \left\langle \dot{A}_0 B_t \right\rangle,$$

the **response function**. (Reference: J. R. Dorfman.)

3 Linear response and discrete-state dynamics

3.1 Electrical conductivity (Green–Kubo)

For $\vec{J} = \sigma \vec{E}$ at frequency ω , $\vec{J}(\omega) e^{i\omega t} = \sigma \vec{E}_0 e^{i\omega t}$. Take N charged particles in a volume V with periodic boundary conditions, $\Gamma = (\vec{r}_1, \dots, \vec{r}_N; \vec{p}_1, \dots, \vec{p}_N)$, and

$$H_0(\Gamma) = \sum_{n=1}^N \frac{|\vec{p}_n|^2}{2m} + \sum_{i<j} \phi_{ij}, \quad H = H_0 - \sum_i q_i \vec{E}(t) \cdot \vec{r}_i = H_0 - \sum_i q_i E(t) x_i \quad (\vec{E} = E(t) \hat{x}).$$

In the language above, $\varepsilon\alpha(t) \rightarrow E(t)$, $A \rightarrow \sum_i q_i x_i$ and $B = J_x = \sum_i q_i \dot{x}_i = \sum_i q_i p_{x,i}/m = \dot{A}$. Then

$$\langle J_x \rangle_t = \int_0^t ds E(s) \beta \langle J_x(0) J_x(t-s) \rangle = \beta \int_0^t dz \langle J_x(0) J_x(z) \rangle E(t-z) \quad (z = t-s).$$

For $E(t) = E_0 e^{i\omega t}$,

$$\langle J_x \rangle_t = \beta \int_0^t dz e^{-i\omega z} \langle J_x(z) J_x(0) \rangle E_0 e^{i\omega t},$$

which for $t \rightarrow \infty$ becomes a Fourier transform:

$$\vec{J}(\omega) = \frac{\langle J_x \rangle}{V} = \underbrace{\frac{\beta}{V} \left[\int_0^\infty dz e^{-i\omega z} \langle J_x(z) J_x(0) \rangle \right]}_{\sigma(\omega)} \vec{E}_0,$$

a Green–Kubo formula for the conductivity. The absorbed power is $P = \vec{E} \cdot \vec{J}V$; with $\vec{E} = E_0 \cos(\omega t)$ its time average is $\bar{P} = \frac{1}{2} E_0 J V$, and the entropy production is $\dot{S} = \beta \bar{P}$.

3.2 Discrete-state dynamics

Applications: biophysics (molecular models, e.g. kinesin carrying cargo along a microtubule), quantum mechanics, surfing the web.

Linear algebra. For a matrix M with right and left eigenvectors

$$M\vec{u} = \lambda\vec{u}, \quad \vec{v}^\top M = \lambda\vec{v}^\top, \quad \vec{v}_m^\top \vec{u}_n = \delta_{mn}$$

(bi-orthonormality), if M is diagonalisable then $M = UDV^\top$, with the right eigenvectors as the columns of U , the eigenvalues on the diagonal of D and the left eigenvectors as the rows of $V^\top$.

Markov jump processes on graphs. States are nodes; allowed transitions are edges. We assume:

1. transitions (jumps) are Poisson processes in continuous time (no memory);
2. there are no directed edges;
3. the graph is connected (no infinite barriers).

A trajectory jumps $i_0 \rightarrow i_1 \rightarrow i_2 \rightarrow \dots$ at times $t_1, t_2, \dots$; in the ensemble view,

$$\vec{p}(t) = (p_1(t), \dots, p_N(t))^\top, \quad p_i(t) = \text{probability of the system being in state } i.$$

Master equation. Normalisation, $\sum_i p_i(t) = 1$, reads $\vec{1}^\top \vec{p}(t) = 1$ with $\vec{1}^\top = (1, \dots, 1)$. For $i \neq j$, $R_{ij} \geq 0$ is the probability rate to jump to state i when in state j (a rate, not a probability). Then

$$\frac{dp_i}{dt} = \sum_{j \neq i} (R_{ij} p_j - R_{ji} p_i).$$

$\sum_{j \neq i} R_{ji}$ is the escape rate from i ; defining $R_{ii} := -\sum_{j \neq i} R_{ji} < 0$,

$$\frac{dp_i}{dt} = \sum_{j \neq i} R_{ij} p_j + R_{ii} p_i = \sum_{j=1}^N R_{ij} p_j \quad \implies \quad \frac{d\vec{p}}{dt} = R\vec{p},$$

with R having strictly negative diagonal and non-negative off-diagonal elements. Columns sum to zero, $\sum_i R_{ij} = \sum_{i \neq j} R_{ij} + R_{jj} = 0$, i.e. $\vec{1}^\top R = \vec{0}^\top = 0 \cdot \vec{1}^\top$, so there exists $\vec{\pi}$ with

$$R\vec{\pi} = \vec{0}, \quad \vec{1}^\top \vec{\pi} = 1,$$

the stationary state ($d\vec{\pi}/dt = R\vec{\pi} = 0$), and $\vec{p}(t) \xrightarrow{t \rightarrow \infty} \vec{\pi}$.

Example: $N = 3$. $\vec{p} = (p_1, p_2, p_3)^\top$ with $p_1 + p_2 + p_3 = 1$, $p_i \geq 0$: the probability vector lives on the triangle (simplex) with vertices $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$. Probability is conserved: $\vec{1}^\top d\vec{p}/dt = \vec{1}^\top R\vec{p} = 0$.

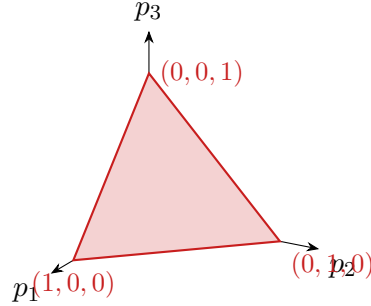


Figure 3.1: The probability simplex for $N = 3$.

Evolution operator. For $\Delta t > 0$,

$$T = e^{R\Delta t} = (e^{R\Delta t/n})^n \quad (\text{a sort of path integral}), \quad T\vec{p}(0) = \vec{p}(\Delta t).$$

T_{ij} is the probability that the system is in state i at time Δt given it was in j at time 0, so $\sum_i T_{ij} = 1$, i.e. $\vec{1}^\top T = \vec{1}^\top$, and $T\vec{\pi} = \vec{\pi}$. Moreover $T_{ij} > 0$ because the graph is connected. Starting from $\vec{p}(0) = (1, 0, 0)^\top$, $\vec{p}(\Delta t) = (T_{11}, T_{21}, T_{31})^\top > 0$ lies strictly inside the triangle. T maps the simplex S into a smaller triangle S' with area ratio $r < 1$, and after n steps the ratio is $r^n \rightarrow 0$: all initial conditions converge to a single point.

Statement (Perron–Frobenius theorem). For positive (real) square matrices, $M_{ij} > 0 \forall i, j$:

- there is a real eigenvalue μ_{PF} strictly greater than the absolute value of all other eigenvalues, $\mu_{\text{PF}} > |\mu_i|$;
- the corresponding left and right eigenvectors are strictly positive, $M\vec{u}_{\text{PF}} = \mu_{\text{PF}}\vec{u}_{\text{PF}}$ with $(u_{\text{PF}})_i > 0$; since $\vec{v}_i^\top \vec{u}_{\text{PF}} = 0$ for the other left eigenvectors, these have positive and negative components;
- $\vec{u}_{\text{PF}}$ and $\vec{v}_{\text{PF}}^\top$ are unique, and μ_{PF} is non-degenerate.

For $T = e^{R\Delta t}$ ($T_{ij} > 0$), $\vec{1}^\top T = \vec{1}^\top$ means $\vec{1}^\top$ is the PF left eigenvector with $\mu_{\text{PF}} = 1$; $T\vec{\pi} = \vec{\pi}$ so all $\pi_i > 0$ (Perron–Frobenius); and all other eigenvalues satisfy $|\mu_i| < 1$. Since $R\vec{u} = \lambda\vec{u} \Rightarrow T\vec{u} = e^{\lambda\Delta t}\vec{u}$, $\mu = e^{\lambda\Delta t}$: $\mu_{\text{PF}} = 1 \Rightarrow \lambda = 0$, and $|\mu_i| < 1 \Rightarrow \text{Re } \lambda_i < 0$, so $\lim_{t \rightarrow \infty} e^{\lambda_i t} = 0$ for all $i \neq \text{PF}$.

If T (hence R) is diagonalisable, $R = U \text{diag}(\lambda_0 = 0, \lambda_1, \dots) V^\top$ with $\text{Re } \lambda_i < 0$ for $i \neq 0$, and

$$\vec{p}(t) = e^{Rt}\vec{p}(0) = U \text{diag}(1, e^{\lambda_1 t}, \dots, e^{\lambda_{N-1} t}) V^\top \vec{p}(0) \xrightarrow{t \rightarrow \infty} \vec{\pi} \underbrace{\vec{1}^\top \vec{p}(0)}_1 = \vec{\pi},$$

since the first column of U is $\vec{\pi}$ and the first row of $V^\top$ is $\vec{1}^\top$. If T is not diagonalisable, its Jordan canonical form has blocks $\mu_i \mathbb{I} + (\text{nilpotent})$ of size $k \times k$ (k the degeneracy of the eigenvalue, broken by the interaction) with $|\mu_i| < 1$, and these still decay, e.g. $\begin{pmatrix} 1/2 & 1 \\ 0 & 1/2 \end{pmatrix}^n \xrightarrow{n \rightarrow \infty} 0$.

Equilibrium vs non-equilibrium stationary states. On a triangle with all rates equal to a ,

$$R = \begin{pmatrix} -2a & a & a \\ a & -2a & a \\ a & a & -2a \end{pmatrix}, \quad \vec{\pi} = \frac{1}{3}(1, 1, 1)^\top,$$

and there is no net current. With clockwise rates a and counter-clockwise rates b , $a > b$,

$$R = \begin{pmatrix} -a-b & b & a \\ a & -a-b & b \\ b & a & -a-b \end{pmatrix} \implies \vec{\pi} = \frac{1}{3}(1, 1, 1)^T,$$

the same stationary distribution, but now there is a current: an equilibrium state versus a **non-equilibrium stationary state**.

3.3 Detailed balance and cycles

The **steady-state current** is

$$J_{ij}^s = R_{ij}\pi_j - R_{ji}\pi_i.$$

In the example, $J_{12}^s = R_{12}\pi_2 - R_{21}\pi_1 = \frac{1}{3}(b - a) < 0$, and $J_{12}^s = J_{23}^s = J_{31}^s = J_{\text{ccw}}^s$ (counter-clockwise current).

Detailed balance. ($a \neq b$ is a violation of detailed balance.) If $J_{ij}^s = 0 \forall i, j$ the system satisfies **detailed balance**; if $J_{ij}^s \neq 0$ for some i, j it violates detailed balance.

Graph vocabulary: a **path** is a sequence of connected nodes and edges beginning and ending in nodes (e.g. $1-2-3-4$, $3-4-3$); a **cycle** is a path that begins and ends on the same node; a **simple path** has no repeating edges or nodes; a **simple cycle** has no repeating edges or nodes except the initial and final ones; a **tree graph** is a connected graph with no simple cycle.

For a rate matrix R on a graph G :

- if G is a tree, detailed balance is satisfied;
- if G is not a tree, detailed balance holds iff, for every cycle $i \rightarrow j \rightarrow k \rightarrow \dots \rightarrow n \rightarrow i$, the **affinity**

$$A_c := \ln \left(\frac{R_{in}R_{nm} \cdots R_{kj}R_{ji}}{R_{ni}R_{mn} \cdots R_{jk}R_{ij}} \right) = 0$$

(**Kolmogorov's** or **Wegscheider's condition**).

If $A_c > 0$ then $J_c > 0$.

Cycle decomposition. For N nodes and E edges the number of fundamental cycles is $C = E - N + 1$; $C = 0$ for a tree. Removing C edges gives a spanning tree; putting back the removed edges one at a time constructs the fundamental cycles. Detailed balance holds iff $A_c = 0$ for every fundamental cycle.

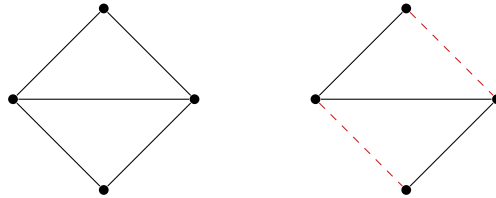


Figure 3.2: A graph with $N = 4$, $E = 5$: $C = 5 - 4 + 1 = 2$ fundamental cycles, obtained by adding back the dashed edges to a spanning tree.

Physics of detailed balance. If the states have energies E_i and the system is at temperature T , the equilibrium state is $\pi_i = \frac{1}{Z} e^{-\beta E_i}$ with $J_{ij}^s = 0 \forall i, j$:

$$R_{ij} \frac{1}{Z} e^{-\beta E_j} - R_{ji} \frac{1}{Z} e^{-\beta E_i} = 0 \implies \frac{R_{ij}}{R_{ji}} = \frac{e^{-\beta E_i}}{e^{-\beta E_j}} = \frac{\pi_i}{\pi_j}.$$

On a triangle, $1 = \frac{\pi_1}{\pi_2} \frac{\pi_2}{\pi_3} \frac{\pi_3}{\pi_1} = \frac{R_{12}}{R_{21}} \frac{R_{23}}{R_{32}} \frac{R_{31}}{R_{13}} = e^{A_c}$.

Relative entropy.

$$\mathcal{D}(\vec{p}|\vec{\pi}) := \sum_{i=1}^N p_i \ln \frac{p_i}{\pi_i}, \quad \frac{d}{dt} \mathcal{D}(\vec{p}|\vec{\pi}) = \sum_i \frac{dp_i}{dt} \ln \frac{p_i}{\pi_i} + \underbrace{\sum_i \dot{p}_i}_0 = \sum_{i,j} R_{ij} p_j \ln \frac{p_i}{\pi_i}.$$

If detailed balance is satisfied,

$$\frac{d}{dt} \mathcal{D}(\vec{p}|\vec{\pi}) = - \sum_{i < j} (R_{ij} p_j - R_{ji} p_i) \ln \frac{R_{ij} p_j}{R_{ji} p_i} \leq 0,$$

since each term has the form $(a - b) \ln(a/b) \geq 0$.

Driven discrete systems. Let the energies depend on a parameter, $E_i(\lambda)$, with rates satisfying $R_{ij}(\lambda)/R_{ji}(\lambda) = e^{-\beta[E_i(\lambda) - E_j(\lambda)]}$, and drive $\lambda : A \rightarrow B$ for $t : 0 \rightarrow \tau$. A trajectory $i_0 \xrightarrow{t_1} i_1 \xrightarrow{t_2} i_2 \rightarrow \dots \rightarrow i_k$ starts in equilibrium $\vec{\pi}_A$. With $E(t) = E_{i(t)}(\lambda(t))$:

- between jumps the energy changes because λ changes: $E(t_1^-) - E(0^+) = E_{i_0}(\lambda(t_1)) - E_{i_0}(\lambda(0))$ is **work**;
- at a jump the energy changes because the state changes: $E(t_1^+) - E(t_1^-) = E_{i_1}(\lambda(t_1)) - E_{i_0}(\lambda(t_1)) = \delta E_1$ is **heat**.

Hence

$$\Delta E = E_{i(\tau)}(\lambda(\tau)) - E_{i(0)}(\lambda(0)) = \sum_k \delta E_k + \int_0^\tau dt \dot{\lambda} \frac{\partial E_{i(t)}}{\partial \lambda} = Q + W,$$

and on average

$$\langle E \rangle = \sum_i p_i(t) E_i(\lambda(t)), \quad \frac{d \langle E \rangle}{dt} = \sum_i \left(\underbrace{\dot{p}_i E_i}_{\langle \dot{Q} \rangle} + \underbrace{p_i \dot{\lambda} \frac{\partial E_i}{\partial \lambda}}_{\langle \dot{W} \rangle} \right).$$

For the relative entropy with respect to the instantaneous equilibrium $\pi_i = \frac{1}{Z(\lambda)} e^{-\beta E_i(\lambda)} = e^{\beta[F(\lambda) - E_i(\lambda)]}$,

$$\dot{\mathcal{D}} = \underbrace{\sum_i \dot{p}_i \ln \frac{p_i}{\pi_i}}_{\leq 0} - \sum_i p_i \frac{d}{dt} \ln \pi_i \leq \beta \sum_i p_i \frac{d}{dt} (E_i(\lambda) - F(\lambda)) = \beta \sum_i p_i \dot{\lambda} \left(\frac{\partial E_i}{\partial \lambda} - \frac{dF}{d\lambda} \right),$$

so $\dot{\mathcal{D}} \leq \beta(\langle \dot{W} \rangle - dF/dt)$ and, since the process starts and ends in equilibrium ($\mathcal{D}(0) = \mathcal{D}(\infty) = 0$),

$$\langle W \rangle \geq \Delta F.$$

3.4 Entropy production and free-energy transduction

For a jump $i \rightarrow j$, $\delta E_{\text{sys}} = E_j - E_i$, $\delta E_{\text{res}} = E_i - E_j$ and

$$\delta S_{\text{res}} = \frac{E_i - E_j}{T} = \ln \frac{e^{-\beta E_j}}{e^{-\beta E_i}} = \ln \frac{R_{ji}}{R_{ij}}.$$

The identification $\delta S_{\text{res}} = \ln(R_{ji}/R_{ij})$ (“local detailed balance”, Seifert 2005) is used either with or without detailed balance. The corresponding change of the system’s (Shannon) entropy is $\delta S_{\text{sys}}^{i \rightarrow j} = -\ln p_j + \ln p_i$.

Example: a three-state heat engine. States $E_1 < E_2 < E_3$; the edge 1–2 is coupled to a cold bath T_c and the edges 2–3, 3–1 to a hot bath $T_h > T_c$:

$$\frac{R_{12}}{R_{21}} = \frac{e^{-E_1/T_c}}{e^{-E_2/T_c}}, \quad \frac{R_{23}}{R_{32}} = \frac{e^{-E_2/T_h}}{e^{-E_3/T_h}}, \quad \frac{R_{31}}{R_{13}} = \frac{e^{-E_3/T_h}}{e^{-E_1/T_h}},$$

$$A_c = \ln \frac{R_{12}R_{23}R_{31}}{R_{21}R_{32}R_{13}} = \frac{E_2 - E_1}{T_c} + \frac{E_3 - E_2}{T_h} + \frac{E_1 - E_3}{T_h} = (E_2 - E_1) \left(\frac{1}{T_c} - \frac{1}{T_h} \right) > 0 :$$

the system cycles, carrying energy $E_2 - E_1$ from the hot to the cold bath.

Example: a chemical reaction. On a triangle where the edge 1–3 converts $A + B \leftrightarrow C + D$,

$$\frac{R_{21}}{R_{12}} = e^{-\beta(E_2 - E_1)}, \quad \frac{R_{32}}{R_{23}} = e^{-\beta(E_3 - E_2)}, \quad \frac{R_{13}}{R_{31}} = e^{-\beta(E_1 - E_3) + \beta X} \neq e^{-\beta(E_1 - E_3)},$$

so

$$A_c = \ln \frac{R_{21}R_{32}R_{13}}{R_{12}R_{23}R_{31}} = \beta X.$$

One cycle converts $C + D \rightarrow A + B$, with $\delta S_{\text{res}}^{C+D \rightarrow A+B} = \beta(\mu_C + \mu_D - \mu_A - \mu_B) = -\beta \Delta G_{\text{rxn}}(C + D \rightarrow A + B)$, so

$$X = -\Delta G_{\text{rxn}} = \mu_C + \mu_D - \mu_A - \mu_B,$$

and if $\mu_C + \mu_D > \mu_A + \mu_B$ then $X > 0$.

Free-energy transduction. On a graph with two cycles coupled to two reactions with $X_1 = \mu_A - \mu_B$ and $X_2 = \mu_C - \mu_D$, the currents oppose one another: if $X_2 \gg X_1 > 0$, the cycle driven by X_2 forces the other cycle to run against its own affinity, transducing free energy.

Birth–death process. With expression rate α and death rate β , $n \in \{0, 1, 2, \dots\}$, $R_{nm} = 0$ if $|n - m| > 1$, $R_{n+1,n} = \alpha$, $R_{n,n+1} = (n+1)\beta$. The graph is a (linear) tree, so detailed balance holds. From $R\vec{\pi} = 0$,

$$\frac{\pi_{n+1}}{\pi_n} = \frac{1}{n+1} \frac{\alpha}{\beta} =: \frac{\kappa}{n+1} \implies \pi_n = \frac{\kappa^n}{n!} e^{-\kappa},$$

a **Poisson distribution** with $\langle n \rangle = \kappa = \alpha/\beta$. For a polymer with attachment rate α and detachment rate β ($R_{n+1,n} = \alpha$, $R_{n,n+1} = \beta$),

$$\pi_n = (1 - \kappa) \kappa^n, \quad \kappa = \frac{\alpha}{\beta} < 1.$$

4 Thermal ratchets and fluctuation theorems

4.1 Thermal ratchets

A ratchet wheel with a pawl is connected by an axle to vanes immersed in a gas. This seems to violate the second law (rectifying thermal fluctuations into work), but if there is just one thermal bath everything is at T_{bath} , in particular the pawl, and every once in a while the wheel rotates in the other direction. If two thermal baths are present, work can be extracted without violating the second law.

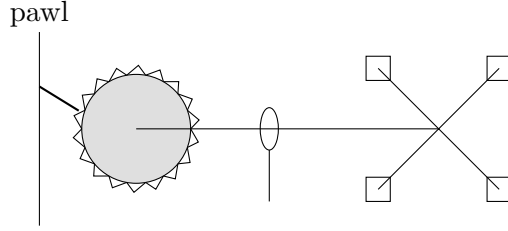


Figure 4.1: Feynman's ratchet and pawl.

Flashing ratchet. A Brownian particle at temperature T in a potential that switches between a flat V_0 and an asymmetric sawtooth $V_1(x)$: when V_1 is on the particle localises at a minimum; when it is off it diffuses symmetrically; when V_1 is switched on again it is more likely to fall into the neighbouring well on the short side. After a while there is a net drift, even if the V_0 potential is slightly tilted against it. The same would happen if V_0 and V_1 were the two states of a system driven by a temperature T' .

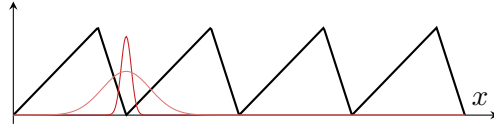


Figure 4.2: Flashing ratchet: sawtooth potential $V_1(x)$, a localised distribution and its spread after the potential is switched off. Net drift to the left.

Metropolis rule.

$$R_{ij} = R(j \rightarrow i) = \Gamma \begin{cases} 1, & E_i < E_j, \\ e^{-\beta(E_i - E_j)}, & E_i > E_j, \end{cases} \quad \frac{R_{ij}}{R_{ji}} = \frac{e^{-\beta E_i}}{e^{-\beta E_j}} \text{ (local detailed balance).}$$

A ratchet between two baths. A rotor coupled to baths at T_A and T_B can be modelled on a graph with $N = 6$ states and $E = 9$ edges, i.e. $C = E - N + 1 = 4$ fundamental cycles. Solving $R\vec{\pi} = 0$ gives the steady currents, e.g. $J_{21}^s = R_{21}\pi_1 - R_{12}\pi_2$ and $J_{54}^s = R_{54}\pi_4 - R_{45}\pi_5$, and the mean velocity $v = d(J_{21}^s + J_{54}^s)$. With $\mu = e^{-\alpha/T_A}$, $\nu = e^{-\alpha/T_B}$ one finds

$$v = -3d \frac{\Gamma}{\mathcal{N}} (\mu - \nu)(1 - \nu)(3\mu + 4), \quad \dot{Q}_{A \rightarrow B} = 3\alpha \frac{\Gamma}{\mathcal{N}} (\mu - \nu)P(\mu, \nu),$$

where $\mathcal{N}(\mu, \nu) > 0$ is a fourth-order polynomial and $P(\mu, \nu) > 0$. The entropy production is

$$\dot{S} = \left(\frac{1}{T_B} - \frac{1}{T_A} \right) \dot{Q} = 3 \frac{\Gamma}{\mathcal{N}} \ln \frac{\mu}{\nu} (\mu - \nu)P(\mu, \nu) \geq 0.$$

With an applied load (a tilted sawtooth of slope f , force $-f$, power $-vf$), the device works as a heat engine or as a refrigerator depending on $T_B - T_A$ and the force.

References: Jarzynski & Mazonka (PRE, 1999); Jülicher, Ajdari & Prost (Rev. Mod. Phys., 1997) for biophysics. The Onsager reciprocal relations earned a Nobel prize.

Fluctuating work in small systems. Stretching a rubber band in air or an RNA molecule in water between optical traps, $\lambda : A \rightarrow B$. For a macroscopic system the work distribution is sharp, $\rho(W) = \delta(W - \bar{W})$ with $\bar{W} \geq \Delta F = F_B - F_A$; for a molecule the work fluctuates on the scale of $k_B T$ (1 pN · nm = 1 zJ, and $k_B T \approx 4$ pN · nm at room temperature), and $\rho(W)$ is broad.

4.2 The Jarzynski equality

The second law holds on average, $\langle -W_R \rangle \leq \Delta F \leq \langle W_F \rangle$, but individual realisations can violate it. Two exact results:

$$\langle e^{-\beta W} \rangle = e^{-\beta \Delta F}, \quad \frac{\rho_F(+W)}{\rho_R(-W)} = e^{\beta(W - \Delta F)},$$

and a stronger statement, $\langle \delta(x - x_t) e^{-\beta W_t} \rangle = \frac{1}{Z_A} e^{-\beta H(x, \lambda_t)}$.

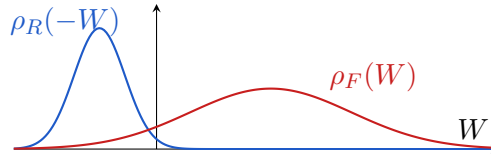


Figure 4.3: Forward and (reflected) reverse work distributions; they cross at $W = \Delta F$.

“Democratic” and “undemocratic” ensembles. The usual density counts every trajectory equally,

$$f(x, t) = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{x_t \in c} 1,$$

where c is a small cell of volume ϵ around x . Another statistical representation of the same ensemble weights each trajectory by its work,

$$g(x, t) = \langle \delta(x - x_t) e^{-\beta W_t} \rangle = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{x_t \in c} e^{-\beta W_t}, \quad W_t = \int_0^t dt' \dot{\lambda}(t') \frac{\partial H}{\partial \lambda}(x(t'), \lambda(t')).$$

Claim: $g(x, t) \propto f^{\text{eq}}(x, \lambda_t)$. Precisely,

$$g(x, t) = \frac{1}{Z_A} e^{-\beta H(x, \lambda_t)} = \frac{Z_{\lambda_t}}{Z_A} p^{\text{eq}}(x, \lambda_t) \xrightarrow{\int dx} \langle e^{-\beta W_t} \rangle = \frac{Z_{\lambda_t}}{Z_A} = e^{-\beta(F_{\lambda_t} - F_A)}.$$

Proof for Hamiltonian dynamics. For $H(x, \lambda)$, $x = (q, p) = (\vec{q}_1, \dots, \vec{p}_N)$: at $t = 0$, $f(x, 0) = \frac{1}{Z_A} e^{-\beta H(x, A)}$; for $0 < t < \tau$, $\lambda(t) : A \rightarrow B$ with $\dot{q} = \partial_p H$, $\dot{p} = -\partial_q H$. Then

$$W = H(x_\tau, \lambda(\tau)) - H(x_0, \lambda(0)) = \int_0^\tau dt \dot{\lambda} \frac{\partial H}{\partial \lambda},$$

and the map $x_0 \mapsto x_\tau = X_\tau(x_0)$ has unit Jacobian, $|\partial x_0 / \partial x_\tau| = 1$ (Liouville's theorem). Hence

$$\begin{aligned}\langle e^{-\beta W} \rangle &= \int dx_0 \frac{1}{Z_A} e^{-\beta H(x_0, A)} e^{-\beta W(x_0)} = \frac{1}{Z_A} \int dx_0 e^{-\beta H(x_\tau(x_0), B)} \\ &= \frac{Z_B}{Z_A} \int dx_\tau \frac{1}{Z_B} e^{-\beta H(x_\tau, B)} \underbrace{\left| \frac{\partial x_\tau}{\partial x_0} \right|^{-1}}_1 = \frac{Z_B}{Z_A}, \\ \langle e^{-\beta W} \rangle &= e^{-\beta \Delta F},\end{aligned}$$

the **fluctuation theorem** (Jarzynski equality).

System coupled to a reservoir. With $H(x, y; \lambda) = H_{\text{sys}}(x; \lambda) + H_{\text{res}}(y) + H_{\text{int}}(x, y)$ and $f(x, y; 0) = \frac{1}{Z_A} e^{-\beta H(x, y; A)}$, the same argument for the isolated system + reservoir gives $W = H(x_\tau, y_\tau; B) - H(x_0, y_0; A)$ and $\langle e^{-\beta W} \rangle = Z_B / Z_A = e^{-\beta \Delta F}$. The equilibrium state of the system is $f_{\text{sys}}^{\text{eq}}(x, \lambda) = \frac{1}{Z_\lambda} e^{-\beta [H_{\text{sys}}(x, \lambda) + \phi(x)]}$ (with the potential of mean force ϕ), $F_\lambda = -\beta^{-1} \ln Z'_\lambda$, and $Z_B / Z_A = Z'_B / Z'_A$.

Quantum version (two-point measurement scheme). With $\hat{H}(\lambda) |n_\lambda\rangle = E_n(\lambda) |n_\lambda\rangle$ and $\hat{\pi}_\lambda = \frac{1}{Z_\lambda} \sum_n e^{-\beta E_n(\lambda)} |n_\lambda\rangle \langle n_\lambda|$, measure the energy at $t = 0$ (outcome $E_n(A)$) and at $t = \tau$ (outcome $E_m(B)$), so $W = E_m(B) - E_n(A)$. With $P(m|n) = |\langle m | \hat{U} | n \rangle|^2$ and unitary evolution,

$$\langle e^{-\beta W} \rangle = \sum_{n,m} \frac{1}{Z_A} e^{-\beta E_n(A)} e^{-\beta (E_m(B) - E_n(A))} P(m|n) = \frac{1}{Z_A} \sum_m e^{-\beta E_m(B)} \underbrace{\sum_n |\langle m | \hat{U} | n \rangle|^2}_1 = \frac{Z_B}{Z_A}.$$

References: C. Jarzynski (PRX, 2017); U. Seifert (PRL, 2016); C. Jarzynski (J. Stat. Mech., 2004) for the simplest derivation; Hummer & Szabo (PNAS, 2001).

4.3 The Crooks fluctuation theorem

Consider $H(x, \lambda)$, $x = (q, p)$, $t : 0 \rightarrow \tau$, with a forward protocol $F : \lambda^F(t) : A \rightarrow B$ and the reverse protocol $R : \lambda^R(t) = \lambda^F(\tau - t) : B \rightarrow A$. Define $x^* = (q, p)^* = (q, -p)$ and assume **time-reversal invariance**, $H(x^*, \lambda) = H(x, \lambda)$ (e.g. $H = \frac{p^2}{2m} + V(q)$).

For a trajectory $X = \{x_t\}$ from x_0 to x_τ , the time-reversed trajectory $X^\dagger$, with $x_t^\dagger = x_{\tau-t}^*$, goes from x_τ^* to x_0^* ; both are solutions of the equations of motion (forward and reverse protocols). For these conjugate pairs $(X, X^\dagger)$,

$$\begin{aligned}W_F[X] &= H(x_\tau, B) - H(x_0, A), & W_R[X^\dagger] &= H(x_0^*, A) - H(x_\tau^*, B) = -W_F[X], \\ P_F[X] &= \frac{1}{Z_A} e^{-\beta H(x_0, A)}, & P_R[X^\dagger] &= \frac{1}{Z_B} e^{-\beta H(x_\tau^*, B)},\end{aligned}$$

since the probability of a deterministic trajectory is that of sampling its initial point from equilibrium. Therefore

$$\frac{P_F[X]}{P_R[X^\dagger]} = \frac{Z_B}{Z_A} e^{\beta (H(x_\tau, B) - H(x_0, A))} = e^{\beta (W_F[X] - \Delta F)}.$$

For the work distributions,

$$\begin{aligned}
\rho_F(W) &= \int dx_0 \frac{1}{Z_A} e^{-\beta H(x_0, A)} \delta(W - H(x_\tau(x_0), B) + H(x_0, A)) \\
&= \frac{Z_B}{Z_A} \int dx_0 \frac{1}{Z_B} e^{-\beta H(x_\tau, B)} e^{\beta[H(x_\tau, B) - H(x_0, A)]} \delta(\dots) \\
&= e^{-\beta \Delta F} e^{\beta W} \int dx_\tau^* \frac{1}{Z_B} e^{-\beta H(x_\tau^*, B)} \delta(W + \underbrace{H(x_0^*, A) - H(x_\tau^*, B)}_{W_R[X^\dagger]}) = e^{\beta(W - \Delta F)} \rho_R(-W), \\
\frac{\rho_F(W)}{\rho_R(-W)} &= e^{\beta(W - \Delta F)},
\end{aligned}$$

the **Crooks fluctuation theorem**; it requires time-reversal symmetry. If the process starts and ends in equilibrium, $\beta(W - \Delta F)$ is proportional to the entropy production. Integrating $\rho_F(W)e^{-\beta W} = \rho_R(-W)e^{-\beta \Delta F}$ over W recovers $\langle e^{-\beta W} \rangle_F = e^{-\beta \Delta F}$ (conceptually simpler).

Proof for overdamped Langevin dynamics. Claim: $g(x, t) := \langle \delta(x - x_t) e^{-\beta W_t} \rangle = \frac{1}{Z_A} e^{-\beta V(x, \lambda_t)}$. With $\partial_t f = \frac{1}{\gamma} \partial_x (\partial_x V f) + D \partial_x^2 f = \hat{\mathcal{L}}_\lambda f$ and $\hat{\mathcal{L}}_\lambda \pi_\lambda = 0$, $\pi_\lambda = \frac{1}{Z_\lambda} e^{-\beta V(x, \lambda)}$, the pair (x, w) obeys

$$\dot{x} = -\frac{1}{\gamma} \frac{\partial V}{\partial x} + \xi(t), \quad \dot{w} = \dot{\lambda} \frac{\partial V}{\partial \lambda},$$

so the joint density $h(x, w, t) = \langle \delta(x - x_t) \delta(w - w_t) \rangle$ satisfies

$$\frac{\partial h}{\partial t} = \hat{\mathcal{L}}_\lambda h - \dot{\lambda} \frac{\partial V}{\partial \lambda} \frac{\partial h}{\partial w}.$$

Then $g(x, t) = \int dw h(x, w, t) e^{-\beta w}$ and, integrating by parts in w ,

$$\frac{\partial g}{\partial t} = \int dw \left[\hat{\mathcal{L}}_\lambda h - \dot{\lambda} \frac{\partial V}{\partial \lambda} \frac{\partial h}{\partial w} \right] e^{-\beta w} = \hat{\mathcal{L}}_\lambda g - \beta \dot{\lambda} \frac{\partial V}{\partial \lambda} g.$$

The candidate $g = \frac{1}{Z_A} e^{-\beta V(x, \lambda_t)}$ satisfies it: $(\hat{\mathcal{L}}_\lambda - \beta \dot{\lambda} \partial_\lambda V) \frac{1}{Z_A} e^{-\beta V} = -\beta \dot{\lambda} \partial_\lambda V \frac{1}{Z_A} e^{-\beta V} = \partial_t (\frac{1}{Z_A} e^{-\beta V})$, with the correct initial condition. ■

The second law from Jensen's inequality. Since $e^y \geq y + 1$, $\langle e^{x - \langle x \rangle} \rangle \geq \langle x - \langle x \rangle + 1 \rangle = 1$, i.e. $\langle e^x \rangle \geq e^{\langle x \rangle}$ (an example of Jensen's inequality). Thus

$$e^{-\beta \Delta F} = \langle e^{-\beta W} \rangle \geq e^{-\beta \langle W \rangle} \implies \langle W \rangle \geq \Delta F.$$

Bound on violations.

$$\begin{aligned}
\text{Prob}(W \leq \Delta F - \epsilon) &= \int_{-\infty}^{\Delta F - \epsilon} dW \rho(W) \leq \int_{-\infty}^{\Delta F - \epsilon} dW \rho(W) e^{\beta(\Delta F - \epsilon - W)} \\
&= e^{\beta(\Delta F - \epsilon)} \int_{-\infty}^{\Delta F - \epsilon} dW \rho(W) e^{-\beta W} \leq e^{\beta(\Delta F - \epsilon)} \int_{-\infty}^{\infty} dW \rho(W) e^{-\beta W} = e^{-\beta \epsilon},
\end{aligned}$$

$$\text{Prob}(W \leq \Delta F - \epsilon) \leq e^{-\beta \epsilon},$$

a microscopic upper bound on violations of the second law. Setting $\epsilon = 0^+$, $\text{Prob}(W \leq \Delta F) < 1$: not all realisations can violate the second law.

Is there a better bound? No. Example: a gas compressed by a thin piston from volume A to B , $\Delta F = -\beta^{-1} \ln(B/A) > 0$; with $B/A \simeq 0.9$, in about 90% of realisations no molecule is hit and $W = 0$, while the rare collisions give large work, so $\langle W \rangle \geq \Delta F$ still holds. Example: free expansion of an ideal gas from V to $2V$ with a piston moving at speed u : $\Delta F = -N\beta^{-1} \ln 2$, yet typically $\langle e^{-\beta W} \rangle = 1 \neq 2^N$. The equality requires the number of realisations $K \rightarrow \infty$ before $u \rightarrow \infty$: it is dominated by rare realisations (e.g. half of the particles having $v \simeq 2u$). (C. Jarzynski, PRE 2006.)

4.4 The arrow of time

(“Arrow of time”, Eddington, 1920s.) From Crooks, $\frac{P_F[X]}{P_R[X^\dagger]} = e^{\beta(W-\Delta F)}$. With Bayes’ rule, $p(a|b)p(b) = p(b|a)p(a)$ (the joint probability of observing a and b), suppose we are shown a movie X and must guess whether it runs forward (F) or backward (R), with priors $P(F) = P(R) = \frac{1}{2}$ and $P(F|X) + P(R|X) = 1$:

$$P(F|X) = \frac{P(X|F)P(F)}{P(X)}, \quad P(R|X) = \frac{P(X|R)P(R)}{P(X)},$$

$$\frac{P(F|X)}{P(R|X)} = \frac{P(X|F)}{P(X|R)} = \frac{P_F[X]}{P_R[X^\dagger]} = e^{\beta(W-\Delta F)},$$

$$P(F|X) = \frac{1}{1 + e^{-\beta(W-\Delta F)}} = 1 - P(R|X).$$

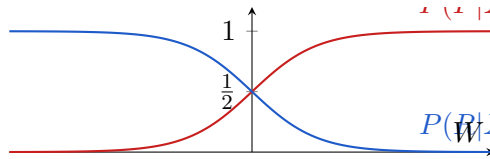


Figure 4.4: Probability that a movie runs forward or backward, as a function of the work performed.

For Gaussian work distributions, $\langle e^{-\beta W} \rangle = e^{-\beta \Delta F}$ forces $\rho_F(W) = \mathcal{N}(\Delta F + \frac{\beta \sigma^2}{2}, \sigma^2)$ and $\rho_R(-W) = \mathcal{N}(\Delta F - \frac{\beta \sigma^2}{2}, \sigma^2)$. For a macroscopic process (work measured in J rather than zJ) the two are far apart and the arrow of time is unambiguous.

Reference: A. Seif, M. Hafezi & C. Jarzynski, *Machine learning the thermodynamic arrow of time*, Nat. Phys. (2021): the logistic form of $P(F|X)$ is the activation function of a neural network. Shannon entropy can be used even out of equilibrium.